WEEK 2 · DAY 3 · RULE BOOK

Lab Day + Quiz 1 + Pick Your Project

Follow it step by step. Every common error and its fix is here.

Today you drive. One investigation from start to finish (**10 checks + 🌶️ bonus**), then Quiz 1, then your final-project kickoff. Fewer hints than usual: use this page plus your rule books and glossaries from Days 1–5.

Before you start (2 minutes)

- ☐ Open `W2D3-exercise` in Colab → **File** → **Save a copy in Drive**. Work in the new tab.
- ☐ Run **STEP 0**. Wait for ✅ `Ready!`
- ☐ Have your Day 1–5 rule books and glossaries open in another tab.

The 6 golden rules of an investigation

#	Rule	Why
1	**Load the right file.** `battery_health_daily` exists in week-03 (clean) AND week-04 (planted faults)	Same name, different data. The wrong one gives the wrong story
2	**Profile before you plot**	Gaps and duplicates silently break everything after
3	**Threshold on the chart, finding in the title**	Your oga reads the title, not the axes
4	**Trailing windows only**	A centred window peeks at the future
5	**The process picks the period**	Monthly data with a yearly calendar → 12 (or half-year → 6)
6	**The correlogram confirms; it doesn't guess for you**	Measure the repeat, don't assume it

The investigation, step by step

Task	What you do	Tool
L1	Choose the folder, time column and rhythm	`"week-03"`, the date column, `"D"`
L2	Rows, empty cells, duplicate timestamps	`len()`, `.isna().sum().sum()`, `.index.duplicated().sum()`
L3	First day capacity drops below 97 Ah	`bat.index[ condition ][0]`

Task	What you do	Tool
L4	Chart with the line + a finding title	a sentence, in quotes
L5	Window (days) that swallows the eclipse dips	about half a year, or a full year
L6	How far the smoothed voltage fell	`smooth.dropna().iloc[0] - smooth.iloc[-1]`
L7	Decomposition period for monthly data	months in a year (half also works)
L8	The two months with the biggest seasonal bump	read the bar chart
L9	Tallest correlogram bar after lag 1	read the plot or the printed list
L10	Your 4-sentence finding	Situation · Observation (with a number!) · Implication · Recommendation
🌶️	When do the two twin files first differ?	subtract the columns, filter, first date

The 4-sentence finding

Sentence	Answers	Must have
Situation	What's going on?	the process and who runs it
Observation	What did the data show?	**a number**
Implication	So what?	the operational meaning
Recommendation	What should we do?	an action someone can take

When things go wrong: error → fix

You see	It means	Fix
`AssertionError: ✍️ Do L1 first…`	A setting above is still blank	Fill it in, run that cell, then this one
`KeyError: 'timestamp'`	This file's time column has a different name	Look at the CSV header: it's `date`
`ValueError: … must have 2 complete cycles`	The period is too big for 36 months	Use a period of 18 or less
`IndexError: index 0 is out of bounds`	Your filter matched nothing	Check the `<` direction and the line value
Smoothed line starts late	Normal: the first `WINDOW - 1` values are `NaN`	Use `.dropna()` before `.iloc[0]`

You see	It means	Fix
`NameError`	Colab forgot	Run STEP 0 and the cells above

Quiz 1 rules (20 minutes)

- ☐ Phones face down, notebooks closed, **no AI tools**.
- ☐ One answer per question. True/False questions have two options.
- ☐ Read the whole question; don't hunt for patterns in the letters.
- ☐ We mark it together. Wrong answers are where you learn the most.

Project kickoff: before you leave today

- ☐ Picked **one** brief (A–E).
- ☐ Named a **process** with an **owner**, not just a topic.
- ☐ Filled in the **project kickoff worksheet**.
- ☐ Sent (or scheduled) the **email to your unit head**.

🏁 **Done with the lab?** Run `scoreboard()`, try the 🌶️ bonus, then help a neighbour with their finding.